THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ratio 45 : 60 in its simplest form is: 1

(a) 4 : 3 **(b)** 3 : 4 **(c)** 3 : 5 **(d)** 4 : 4

2. The ratio 25 : 100 in its simplest form is: 1

(a) 1 : 5 **(b)** 1 : 4 **(c)** 4 : 1 **(d)** 2 : 4

3. If 5 : 6 = 15 : x, then x equals: 1

(a) 19 **(b)** 17 **(c)** 18 **(d)** 21

4. ₹ 900 is divided between A and B in the ratio 7 : 3. A's share is: 1

(a) ₹ 630 **(b)** ₹ 270 **(c)** ₹ 720 **(d)** ₹ 450

5. 3 hours in minutes is: 1

(a) 30 minutes **(b)** 180 minutes **(c)** 300 minutes **(d)** 360 minutes

6. 90 minutes in hours is: 1

(a) 9 hours **(b)** 15 hours **(c)** 0.9 hours **(d)** 1.5 hours

7. Two ratios are equal if they: 1

(a) have the same value when simplified **(b)** have the same first term
(c) have the same sum **(d)** have equal differences

8. A mixture has milk and water in the ratio 3 : 2. In 20 L of mixture, milk is: 1

(a) 15 L **(b)** 8 L **(c)** 12 L **(d)** 10 L

9. If the ratio of boys to girls is 5 : 4 and there are 36 girls, the number of boys is: 1

(a) 45 **(b)** 36 **(c)** 40 **(d)** 54

10. The ratios 2 : 5 and 6 : x are equal if x is: 1

(a) 18 **(b)** 12 **(c)** 30 **(d)** 15

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Convert 3.6 km to metres and 450 mL to litres. 2

12. If 8 workers finish a job in 15 days, how many days will 12 workers take? 2

13. Find x if $5 : 8 = x : 40$. 2

14. The ratio of cats to dogs in a shelter is 4 : 5. If there are 20 cats, how many dogs? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Two numbers are in the ratio 3 : 2 and their sum is 40. Find the numbers. 3

16. Two numbers are in the ratio 5 : 3 and their sum is 64. Find the numbers. 3

17. Two numbers are in the ratio 4 : 5 and their sum is 72. Find the numbers. 3

OR

Two numbers are in the ratio 7 : 3 and their sum is 50. Find the numbers. 3

18. Two numbers are in the ratio 2 : 7 and their sum is 90. Find the numbers. 3

OR

Two numbers are in the ratio 3 : 8 and their sum is 66. Find the numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) 12 men can build a wall in 20 days. How many days will 15 men take?
(ii) How many men are needed to finish in 8 days? 5

OR

Case study: A school's 480 students are in the ratio 5 : 3 for Primary and Secondary sections. 5
(i) How many in each?
(ii) If 20 Primary students shift to Secondary, find the new ratio.
(iii) Comment.

20. (i) If $A : B = 3 : 4$ and $B : C = 6 : 7$, find $A : B : C$. 5
(ii) If the sum is 175, find each.

OR

(i) Convert 36 km/h to m/s. 5
(ii) A cyclist travels 15 km in 50 minutes; find speed in km/h.

END OF QUESTION PAPER — Total: 40 Marks

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